Inventing Climate Finance. From Incremental Costs to Strategic Ambiguity

Inventer la finance climat. Des coûts incrémentaux à l'ambiguïté stratégique

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Résumé. Comment la « finance climat » est-elle devenue un objet économique à part entière ? Cet article retrace l'histoire de sa construction entre 1990 et 2024, en montrant que les économistes n'ont pas découvert une réalité préexistante mais ont fabriqué les catégories comptables — marqueurs de Rio, méthodologies de finance privée mobilisée, seuils de concessionalité — à travers lesquelles les flux financiers liés au climat ont été rendus mesurables et gouvernables. Une analyse computationnelle d'environ 30,000 travaux corrobore une périodisation en trois actes : une préhistoire où le concept n'existait pas encore (1990–2006), une cristallisation déclenchée par l'engagement de Copenhague (2007–2014), et une période de disputes menées dans les catégories établies lors de la cristallisation (2015–2024). Une tension récurrente entre pôle d'efficacité et pôle de redevabilité traverse le champ. La finance climat est devenue gouvernable non pas malgré mais grâce à l'ambiguïté délibérée de ses catégories comptables — une ambiguïté qui n'était pas un défaut à corriger mais une ressource politique exploitée par toutes les parties.

Abstract. How did "climate finance" become a distinct economic object? This paper traces the intellectual history of its construction from 1990 to 2024, arguing that economists did not discover a pre-existing reality but built the accounting categories — Rio markers, mobilised-private-finance methodologies, concessionality thresholds — through which climate-related financial flows were rendered measurable and governable. A computational analysis of approximately 30,000 works corroborates a three-act periodization: a prehistory when the concept did not yet exist (1990–2006), a crystallization triggered by the Copenhagen commitment (2007–2014), and a period of disputes fought within the categories established during crystallization (2015–2024). Throughout, a recurring tension between an efficiency pole and an accountability pole runs through the field. Climate finance became governable not despite but through the deliberate ambiguity of its accounting categories — an ambiguity that was not a deficiency to be repaired but a political resource exploited by all sides.

Mots-clés : finance climat, quantification, catégories comptables, organisations internationales

Keywords: climate finance, quantification, accounting categories, international organisations

JEL codes: B20, Q54, F35, Q56

In November 2024, negotiators at the Baku climate summit agreed that developed countries should mobilise $300 billion per year for climate action in the developing world. No institution holds this sum. No ledger records it. No methodology for counting it commands consensus. The number was produced—like its $100 billion predecessor pledged at Copenhagen in 2009—by applying accounting conventions that are themselves the object of bitter dispute to heterogeneous financial flows that no single actor controls. How did something so consequential come to rest on categories so contested? Climate finance had to be *made* into a countable object before anyone could argue about whether enough of it existed. This paper asks how economists performed that making.

This paper offers a history-of-economic-thought analysis of the emergence of climate finance as a distinct economic object from the early 1990s to the mid-2020s (Figure 1). It examines how economists actively shaped the categories, metrics, and boundaries through which climate-related financial flows were defined, stabilised, and contested, contributing to the history of climate economics by shifting attention from models to accounting conventions.

The existing historiography of climate economics has focused primarily on modelling. Pottier (2016) provides a critical intellectual history of the cost-benefit tradition from Nordhaus to the Stern Review, showing how the framing of climate change as an optimisation problem marginalised alternative approaches. Aykut and Dalmedico (2015) trace the co-production of climate science and international governance, emphasising how models became political instruments. More broadly, intellectual histories of environmental economics have traced the lineage from Pigou through Coase to cap-and-trade, emphasising the construction of pricing instruments. The parallel construction of *accounting* instruments for tracking financial flows has received no comparable historiographical attention. These works illuminate how economists framed climate change as a *problem*; they say little about how economists made climate finance into a *measurable object*. The shift from "what should the carbon price be?" to "does this dollar count as climate finance?" involves a different kind of economic work—not modelling but accounting, not optimisation but commensuration (Espeland and Stevens, 1998). A growing political economy literature documents the power asymmetries and distributional conflicts embedded in climate finance accounting (Roberts and Weikmans, 2017; Skovgaard, 2017). These works reveal *what is at stake* in measurement disputes but treat the accounting categories themselves as given. We ask the prior question: how were these categories constructed, by whom, and through what intellectual operations? Our contribution addresses this gap: the intellectual history of the accounting conventions and measurement disputes through which climate finance was constituted as a governable economic domain.

Rather than treating climate finance as a technical extension of environmental or development economics, we situate it within a longer trajectory of climate economics. Early work on externalities and collective environmental constraints (Ayres and Kneese, 1969) preceded, and only partially informed, later modelling efforts centred on growth, damages, and risk (Manne and Richels, 1992; Nordhaus, 1992; Stern, 2007; Weitzman, 2007). The burden-sharing *framework* embedded in these models [Negishi weights; Negishi (1960)] remained disconnected from parallel debates on aid and international responsibility, even as some of the same economists

contributed to both literatures. Climate finance emerged through entanglement with international negotiations, development finance practices, and attempts to translate climate commitments into financially actionable terms—rather than as a direct outgrowth of integrated assessment modelling.

At the juncture where climate policy met development finance, economists in international organisations played a decisive role. At the OECD, figures such as Jan Corfee-Morlot built statistical infrastructures—notably the Rio markers—that rendered climate finance measurable within development finance frameworks, defining what could legitimately count and how responsibility could be claimed (Corfee-Morlot et al., 2009, 2012). These accounting categories became focal points of a deeper divide. On one side, economists working within a market-failure framework emphasised efficiency, leverage, and capital mobilisation, framing public climate finance as a catalytic instrument to crowd in private investment (Stern, 2009). On the other, political-economy-oriented economists challenged both measurement conventions and normative assumptions: Michaelowa and Michaelowa (2007) on incentive-driven over-reporting, Roberts and Weikmans (2017; see also Roberts et al., 2021) on accounting disputes as distributive conflicts, Kaul et al. (2003; see also Kaul, 2017) on the distinct financing requirements of global public goods. We trace this divide through four recurrent controversies: the valuation of concessional loans, the credibility of Rio markers, the attribution of mobilised private finance, and the boundary between development assistance and climate-specific obligations.

The paper is organised into three acts. Section 1 traces the intellectual prehistory before climate finance existed as an autonomous object (1990–2006). Section 2 analyses the crystallization triggered by the Copenhagen commitment and the construction of OECD statistical infrastructure (2007–2014). Section 3 shows that all subsequent disputes—the $100 billion accounting battles, the $300 billion NCQG at Baku—have been fought within the stable categories established during crystallization (2015–2024). The conclusion traces how the precise legal framework of Article 4.3 of the UNFCCC was progressively abandoned in favour of a deliberately ambiguous category, arguing that this strategic indeterminacy, rather than theoretical coherence, is what made climate finance governable.

We ground this historical argument in a computational analysis of approximately 30,000 works on climate finance (1990–2024), drawn from six bibliographic sources including OpenAlex, ISTEX, and curated grey literature (OECD, World Bank, UNFCCC reports). Using sentence-transformer embeddings, co-citation analysis, and unsupervised clustering, we find that the corpus exhibits patterns consistent with this periodization — the computational patterns corroborate the historical narrative rather than generating it. The periodization and argument rest on the institutional record: primary sources, policy documents, and the intellectual trajectories of named actors. The computational patterns serve as corroborative signals that discipline the narrative. Each of the three historical sections closes with a "corpus evidence" subsection presenting the relevant quantitative findings; these can be read together as a self-contained empirical companion or skipped without loss of the historical argument. Technical

details of the embedding pipeline, clustering, and classification procedures are documented in a companion paper and replication repository.

A reflexive note is warranted. Our own method, transforming heterogeneous texts into vector representations and detecting statistical patterns, is itself an act of commensuration in Espeland and Stevens (1998)'s sense. The embedding space renders comparable what was written in different languages, disciplines, and institutional contexts. We treat its outputs as one more measurement apparatus whose conventions shape what it makes visible, not as objective descriptions of an independent intellectual landscape.

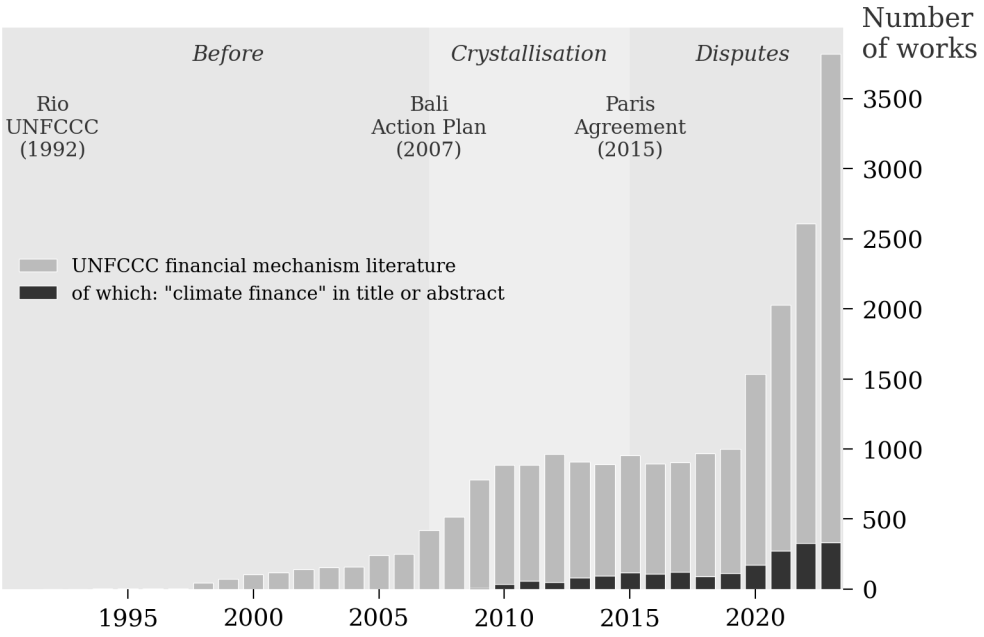

Figure 1: Growth of climate finance scholarship, 1990–2023. The field was negligible *Before* (1990–2006), surged after the Copenhagen 2009 conference during the *Crystallisation* period (2007–2014), and expanded further in the *Disputes* period (2015–2023).

1. Before Climate Finance (1990–2006)

In 1992, the United Nations Framework Convention on Climate Change laid down a financial obligation in precise legal language:

> The developed country Parties [...] shall provide new and additional financial resources to meet the agreed full costs incurred by developing country Parties [...]. They shall also provide such financial resources, including for the transfer of technology, needed by the developing country Parties to meet the agreed full incremental costs of implementing measures [...] agreed between a developing country Party and the international entity or entities referred to in Article 11. (United Nations, 1992, Article 4.3)

The wording was carefully transactional: reimbursement of "agreed full incremental costs" through a bilateral mechanism, not aggregate investment flows. The word "finance" did not appear. The "developed" and "developing" categories themselves were political constructions: the Convention's Annexes reflected 1990 negotiating blocs, not the World Bank's income thresholds or any other economic taxonomy. No mechanism existed for reclassification; no country has moved from non-Annex I to Annex I since 1992. Yet this single paragraph—built on a frozen political binary—established the principle that rich countries owe money to poor countries for the costs of addressing climate change, generating the most persistent accounting disputes of the next three decades. How did that principle become "climate finance"? Not all at once. The term did not exist as an autonomous economic object before the late 2000s; it appears sporadically in the literature of the 1990s and early 2000s but referred to no stable category. What existed instead were three distinct intellectual traditions, each of which would later contribute elements to the concept without any of them producing it on their own.

1.1 Environmental economics and the externality framework

The first tradition is environmental economics, rooted in the treatment of pollution as an externality. From Pigou's welfare economics onward, the core analytical move was to identify divergences between private and social costs and to design corrective instruments (taxes, subsidies, tradable permits). Ayres and Kneese (1969) extended this framework to "materials balance," treating environmental degradation as a systemic by-product of production and consumption rather than an isolated market failure. The subsequent debate between Pigouvian taxation and Coasean bargaining generated a rich literature on instrument design, culminating in the cap-and-trade systems of the 1990 US Clean Air Act and, later, the EU Emissions Trading System. But this tradition did not produce a concept of climate *finance*. Its central question was the *price* of emissions (what should a carbon tax be?), not the *flow* of money between countries to address climate change. Most economists reason about climate policy through constrained optimisation: maximise welfare subject to an emissions ceiling. A standard result of such models is that relaxing a constraint can only improve the objective.

Applied to international negotiations, this became the doctrine of "when and where flexibility": mitigation should occur wherever marginal abatement costs are lowest, and those costs were presumed lower in developing countries. This reasoning produced the Kyoto Protocol's Clean Development Mechanism, which channelled investment from industrialised to developing countries. Yet the framing remained one of efficiency gains from trade, not of financial transfers owed or resources mobilised.

When climate economics emerged as a distinct subfield in the late 1980s and 1990s, it inherited this orientation. The integrated assessment models developed by Nordhaus (1992) and Manne and Richels (1992) framed climate change as an intertemporal optimisation problem: how much should the present generation invest in mitigation to reduce future damages? These models calculated optimal abatement paths and shadow prices of carbon, but they operated at a level of abstraction that left the institutional mechanisms of financing unspecified. Money flowed between "regions" in the models, but these flows were artefacts of welfare optimisation, not representations of actual financial instruments or aid commitments.

The Stern Review (Stern, 2007) marked a turning point in public visibility but not in conceptual architecture. Stern's argument that the benefits of early action outweighed the costs depended on a low discount rate—a choice contested by Weitzman (2007; see also Weitzman, 2009) and others—but the *kind* of action Stern envisaged was still framed in terms of carbon pricing and public investment, not in terms of a distinct "climate finance" category requiring its own accounting infrastructure. The famous "1% of GDP" figure was an aggregate cost estimate, not a financial flow from North to South.

1.2 Development economics and the categories of aid

The second tradition is development economics, which contributed the institutional categories through which climate finance would eventually be counted. The OECD's Development Assistance Committee had been tracking Official Development Assistance (ODA) since the 1960s, building an elaborate statistical system for classifying financial flows by type (grants, loans, equity), channel (bilateral, multilateral), sector, and purpose. This system embodied specific conceptual commitments: the distinction between "concessional" and "non-concessional" flows, the requirement that ODA have a "grant element" of at least 25%, the counting of face value rather than subsidy content. These were not natural categories but historical constructs, stabilised through decades of negotiation among donor countries and crystallised in the DAC statistical directives. As Desrosières (1998) argues, such statistical conventions acquire the force of objectivity once their conventional origins are forgotten.

The DAC system also encoded a particular geography of responsibility: aid flowed from North to South, measured by donor effort rather than recipient need. The 0.7% of GNI target, adopted by the UN General Assembly in 1970, established the metric through which donor generosity would be judged. This framework—bilateral flows, concessionality thresholds, donor-centred accounting—would later be grafted onto climate finance almost unchanged, importing

both its analytical power and its political biases. (The ODA system was not the only precedent: the EU's Structural and Cohesion Funds had been redistributing resources across member states since 1989 using GDP-per-capita thresholds, co-financing rules, and Commission audits. When climate earmarking was imposed on cohesion funds in 2021, Europe reproduced the same classification problem that Rio markers had created globally.)

These categories were designed for development aid, not for climate policy. They constituted, in Desrosières's terms, a pre-existing *statistical convention*—a system of commensuration that rendered heterogeneous flows comparable by absorbing their qualitative differences into standardised metrics (Merry, 2016). But when the need arose to count climate finance after Copenhagen, they provided the readymade infrastructure. The Rio markers, introduced in 1998 to tag aid projects targeting the objectives of the Rio Earth Summit conventions (climate, biodiversity, desertification), were the key link. They allowed development finance statisticians to retrofit climate labels onto existing aid data without building a new reporting system from scratch. This was efficient but consequential: it meant that the way climate finance was counted would reflect the assumptions, compromises, and blind spots of the ODA system (Michaelowa and Michaelowa, 2007). The markers were self-assessed by donors with no external verification—a design choice that would later become the subject of fierce controversy over inflation and credibility.

1.3 Burden-sharing and the disconnection from finance

The third tradition is the economics of burden-sharing in international environmental agreements. Models incorporating Negishi welfare weights (Negishi, 1960) had been used since the 1970s to analyse how the costs of collective action should be distributed among countries with different income levels. The climate negotiations of the 1990s—the Framework Convention (1992), the Kyoto Protocol (1997)—were deeply shaped by this logic: differentiated responsibilities, equity principles, historical emissions. Article 4.3 of the UNFCCC, quoted at the opening of this section, encoded the burden-sharing principle in transactional language: bilateral reimbursement of agreed incremental costs, not aggregate financial flows. This rigorous incremental-cost framework was available when the need for a broader category arose, but it was ultimately abandoned rather than extended. Yet the burden-sharing literature remained disconnected from the practical questions of financial flows. It addressed the question "who should pay?" at a theoretical level but did not engage with the operational question "how do we count what has been paid?" The economic models that informed the negotiations treated transfers between regions as welfare-theoretic constructs, not as real-world financial flows requiring institutional tracking.

The Kyoto Protocol's Clean Development Mechanism (CDM) came closest to bridging this gap. By allowing developed countries to meet emission targets through projects in developing countries, the CDM created actual financial flows tied to climate objectives. A substantial literature emerged around CDM project design, additionality criteria, and carbon credit markets (Michaelowa et al., 2019). The CDM was economically significant—by 2015, over 7,500

projects had been registered in 105 developing countries, nominally attracting over $200 billion in investment according to the UNFCCC Secretariat—though the environmental integrity and additionality of many projects remained contested (Michaelowa et al., 2019). It also introduced the concept of "additionality" as a practical test: would the project have happened without the CDM incentive? This concept would later migrate into climate finance debates, where it was repurposed to ask whether finance was "additional" to existing aid.

But the CDM operated through market mechanisms, not through budgetary transfers, and its financial flows were counted in terms of Certified Emission Reductions (CERs) rather than dollars of "climate finance." The CDM produced a vocabulary of carbon accounting, not of financial accounting. Moreover, the CDM's collapse after 2012—when CER prices fell below €1 as demand evaporated—demonstrated the fragility of market-based transfer mechanisms and reinforced the demand for a more rigorous, budget-based accounting of climate finance.

1.4 Why climate finance did not yet exist

The three traditions described above (environmental economics, development economics, and burden-sharing analysis) each addressed aspects of what would later be called climate finance. Environmental economists modelled the costs of climate action; development economists built the statistical infrastructure for tracking aid flows; burden-sharing analysts debated who should bear the costs. But none of them constituted climate finance as a *distinct economic object*—a category with its own metrics, its own institutions, and its own controversies.

In retrospect, these three traditions had remarkably little contact before the late 2000s. The integrated assessment modellers rarely engaged with the ODA literature; the DAC statisticians had no reason to attend to welfare economics debates about discount rates; the burden-sharing theorists operated at a level of abstraction that made the institutional details of aid delivery invisible. Climate finance as a concept required precisely the *intersection* of these traditions—the moment when a political commitment (burden-sharing), a measurement infrastructure (development economics), and a market-failure logic (environmental economics) were simultaneously demanded. That moment was Copenhagen, and the assembly it triggered is the subject of the next section.

1.5 Corpus evidence

Co-citation analysis confirms that these three traditions operated without meaningful cross-fertilisation before 2007. Applying Louvain community detection to the 250 most frequently cited pre-2007 references in our corpus yields 10 communities (modularity = 0.68) that recover three distinct clusters (see Table 1). The three communities share almost no edges in the co-citation network—empirical evidence that researchers in each tradition were reading, and being read by, distinct audiences. Meanwhile, publication volume was negligible: fewer than 50

works per year before 2007 (Figure 1), spread across communities with no shared institutional home.

The corpus also reveals a gap between intellectual-history narrative and citation practice. The burden-sharing tradition described in §1.3 is anchored in our narrative by Negishi (1960), whose welfare weights informed integrated assessment models by Nordhaus (1992) and others. Yet Negishi (1960) does not appear among the top-250 pre-2007 references in the corpus: climate finance scholars did not cite the welfare-weights framework directly. Nordhaus, Barrett, Tol, and Fankhauser do appear, but they are cited for cost-benefit analysis and environmental agreement design, not for the burden-sharing calculus as such. The specific question "how should costs be distributed?"—operationalised through Negishi weights and equity criteria—left no distinct co-citation footprint. What the algorithm detects as the third pre-2007 community is instead anchored by North (1990), DiMaggio and Powell (1983), and Finnemore and Sikkink (1998)—the institutionalist and constructivist literature that framed the UNFCCC as a site of norm diffusion rather than welfare optimisation. The burden-sharing *question* persisted, but the analytical framework through which climate finance scholars addressed it shifted from welfare economics to institutional analysis.

Table 1: Three intellectual traditions preceding climate finance (pre-2007). *Intellectual-history anchors* are works foregrounded in §1.1–1.3; *co-citation anchors* are the most-connected papers in the algorithmically detected community.

Tradition and core question	Intellectual-history anchors	Co-citation anchors (pre-2007)	Contribution to climate finance
Environmental economics — What price for carbon?	Nordhaus (1992); Manne and Richels (1992); Stern (2007); Weitzman (2007)	Weitzman (1974); Barrett (1994); Carraro and Siniscalco (1993)	Cost–benefit logic, optimal abatement paths, shadow prices of carbon — but no concept of North–South financial *flows*
Development economics — How to count and channel aid?	Desrosières (1998); Michaelowa and Michaelowa (2007); OECD DAC directives; Rio markers (1998)	Michaelowa and Jotzo (2005); Sutter and Parreño (2007)	ODA accounting infrastructure, concessionality thresholds, bilateral/multilateral classifications — the readymade measurement apparatus grafted onto climate finance after Copenhagen
International governance — Through what institutions, by what norms?	Negishi (1960); UNFCCC Art. 4.3 (1992); Kyoto Protocol (1997)	North (1990); DiMaggio and Powell (1983); Finnemore and Sikkink (1998)	Differentiated responsibilities, norm diffusion, "new and additional" resources — governance framing displaced the welfare-theoretic calculus

2. Crystallization (2007–2014)

2.1 The Copenhagen moment

In December 2007, the Bali Action Plan broke with the Convention's incremental-cost logic. Where Article 4.3 had specified bilateral reimbursement of agreed costs, Bali called for something broader:

> Enhanced action on the provision of financial resources and investment to support action on mitigation and adaptation and technology cooperation [...] improved access to adequate, predictable and sustainable financial resources [...] including official and concessional funding for developing country Parties. (UNFCCC, 2007, Decision 1/CP.13, para. 1(e))

The demand that these commitments be "measurable, reportable and verifiable" (MRV) was the first explicit requirement that climate finance be rendered *countable* as a condition of political legitimacy. The Stern Review, published the year before, had already reframed climate change from an environmental externality into a financial risk of historic proportions (Stern, 2007). Together, these interventions created the conditions for a new kind of economic object: one that required not just models but accounting infrastructure.

The decisive moment came at Copenhagen (COP15) in December 2009, when developed countries committed to mobilising $100 billion per year in climate finance by 2020. The commitment was performative in Callon's (1998) sense: a declaration that, by triggering the construction of measurement infrastructure, brought into existence the very object it promised to deliver. This was institutional performativity, distinct from the market-based Barnesian form analysed by MacKenzie (2006). Before Copenhagen, "climate finance" referred loosely to various financial flows related to climate policy—carbon market revenues, GEF grants, bilateral aid tagged with environmental objectives. After Copenhagen, it became a specific, politically binding quantity that had to be defined, measured, tracked, and reported. The $100 billion figure was not derived from any economic model or needs assessment; it emerged from last-minute diplomatic bargaining. The pledge broke definitively with Article 4.3's incremental-cost logic: it specified neither which costs were covered nor what "climate finance" meant—only that $100 billion must be mobilised. Yet precisely because it was arbitrary, it *required* economists to build the measurement apparatus that would give it substance. The methodologies developed to *count* the $100 billion were themselves performative devices: they defined which flows could be claimed as "climate," thereby redirecting investment decisions, reporting systems, and national climate plans. The target performed the reality it purported to measure—imperfectly, contested, yet consequentially.

2.2 Building statistical infrastructure

The economists who stepped into this role were not academic theorists but policy professionals embedded in international organisations—a distinctive case of what Çalışkan and Callon (2009; see also Çalışkan and Callon, 2010) call *economization*, the process by which a domain is constituted as "economic" through the construction of calculative devices. The most consequential infrastructure was already under construction before Copenhagen. At the OECD's Development Assistance Committee (DAC), a system of "Rio markers" had been introduced in 1998 to flag aid projects with climate-relevant objectives. Originally a simple binary tag, the markers were redesigned in the mid-2000s into a three-tier system (principal, significant, not targeted) that became the *de facto* standard for counting bilateral climate finance. Jan Corfee-Morlot, an economist at the OECD's Environment Directorate, played a central role in connecting climate policy analysis with development finance statistics, co-authoring a series of working papers that established the conceptual framework for monitoring climate-related financial flows (Büchner et al., 2011; Corfee-Morlot et al., 2009, 2012). Yet the redesign embedded a critical ambiguity. While the original binary markers served as rough tracking tools, the three-tier system demanded finer judgments without providing the calibration to make them consistent. No agreed coefficient was established for projects marked "significant": individual donor agencies applied anywhere from 0% to 100% of a project's value as their climate finance contribution, making aggregate figures structurally incomparable across countries. Japan and the United Kingdom, reporting on similar projects, could produce different climate finance totals simply through different interpretation of the same marker category.

When the Bali Action Plan (2007) demanded "measurable, reportable and verifiable" financial commitments, the crude binary markers became politically inadequate for accountability. The gap between the precision demanded by MRV and the discretion built into the marker system illustrates a tension at the heart of commensuration: the same categories that rendered climate finance *countable* also made it *contestable*, because the mapping from qualitative reality to quantitative indicator was never fully determinate (Michaelowa and Michaelowa, 2007).

What made this infrastructure consequential was not its technical sophistication but its *institutional authority*. The DAC reporting system was already the standard through which donor countries reported Official Development Assistance. By grafting climate markers onto this existing apparatus, the OECD ensured that climate finance would be counted using development economics categories—grants, loans, concessionality, bilateral versus multilateral channels. This was not a neutral technical choice. It meant that climate finance inherited the conceptual architecture of aid, including its most contested assumptions about what counts as a financial "flow" and who gets credit for it (Desrosières, 1998; Porter, 1995).

A parallel infrastructure emerged within the UNFCCC process, where the Standing Committee on Finance (SCF), established at COP16 in Cancún (2010), was tasked with producing biennial assessments of climate finance flows. The SCF drew on different data sources and operated with different institutional incentives than the OECD. Where the DAC system was designed by and for donors, reporting what they had *provided*, the UNFCCC process was also

accountable to recipient countries, who reported what they had *received*—and the two figures rarely agreed. The SCF's Biennial Assessments attempted to bridge these perspectives, but the exercise revealed how profoundly the choice of data source, boundary definition, and aggregation method shaped the resulting numbers. The first Biennial Assessment (2014) estimated total climate finance flows at $340–$650 billion per year—a range so wide that it illustrated the problem more than it resolved it.

This created a structural tension: two competing measurement regimes, each embedded in different political constituencies, producing different numbers from the same underlying reality. The OECD could draw on decades of established DAC reporting infrastructure; the UNFCCC had the political legitimacy of a multilateral treaty process. The "battle for the operational definition," as we term it, was a struggle over who would control the categories through which climate finance could be known and evaluated. In practice, the OECD's numbers became the *de facto* reference for assessing progress toward the $100 billion target, largely because the DAC system produced data that was more standardised and more timely. This institutional advantage meant that the development-economics categories embedded in the DAC system effectively won the definitional contest—not through intellectual superiority but through infrastructural incumbency. What economization adds to commensuration is a claim about the *kind* of governance that measurement enables: commensuration makes things comparable; economization makes them amenable to financial reasoning—cost-benefit assessment, leverage calculation, de-risking ratios—thereby constituting the climate problem as economic and foreclosing framings centred on reparative justice or ecological debt (Espeland and Stevens, 1998; Merry, 2016).

2.3 The emergence of key concepts

The crystallization period saw the rapid proliferation of a specialised vocabulary. Before 2007, the climate finance literature was dominated by terms inherited from the Kyoto Protocol—CDM, carbon credits, emission trading. After Copenhagen, a new lexicon emerged: "mobilised private finance," "leverage ratios," "blended finance," "de-risking," "crowding-in." Each of these terms encoded specific assumptions about the relationship between public and private capital. "Blended finance," for instance, implied that public and private capital were complementary ingredients that could be mixed in optimal proportions—a metaphor drawn from financial engineering that naturalised the idea that public climate finance should serve primarily as a catalyst for private investment. "De-risking" similarly framed the problem as one of market failure (investors perceive too much risk in climate-relevant investments) rather than development need (countries lack resources for adaptation). The vocabulary was not descriptively neutral; it enacted a particular vision of what climate finance was *for*.

The concept of "mobilised private finance" was particularly consequential. It allowed donor countries to count private investment triggered by public interventions as part of their climate finance contributions, vastly expanding the measurable total. But it also raised fundamental questions about *attribution*: if a development bank provides a partial guarantee for a wind

farm, how much of the total project cost counts as "mobilised"? The OECD convened expert groups to develop standardised methodologies, producing a series of reports that effectively set the terms of debate (Caruso and Ellis, 2013; Jachnik et al., 2015). These were not academic publications in the conventional sense—they were institutional documents that acquired authority through the OECD's convening power and their adoption in DAC reporting guidelines.

The institutional mechanism through which these methodologies were built deserves attention. The OECD's Research Collaborative on Tracking Private Climate Finance (2013–2015), coordinated by Raphaël Jachnik and Roberta Caruso of the Environment Directorate, convened development bank representatives and bilateral agencies to resolve the attribution question through committee work. The group confronted a series of binary methodological choices, each with distributional consequences: volume-based attribution (counting the total project value) favoured large multilateral development bank co-financing arrangements, while instrument-based attribution (counting only finance directly linked to specific public instruments) favoured bilateral guarantees and risk-sharing facilities. Similarly, counting at point of commitment produced larger figures than counting at point of disbursement; counting indirect mobilisation expanded the total beyond what direct causal chains could support. The resulting methodology, adopted into DAC reporting guidelines, illustrates economization through bureaucratic procedure: choices made by a small expert group became binding statistical conventions that determined which countries could claim credit for mobilising private capital (Caruso and Ellis, 2013; Jachnik et al., 2015).

At the same time, concepts from the accountability tradition gained traction. "Additionality," originally referring to emission reductions beyond a business-as-usual baseline in environmental economics, was repurposed to ask whether climate finance was "new and additional" to existing aid commitments, as the Copenhagen Accord had promised. Stadelmann et al. (2011) systematically assessed eight baseline options for operationalising additionality, concluding that only two met both feasibility and equity criteria. The term "double counting" (the risk that the same financial flow might be reported by multiple actors) became a focal point of civil society critique. And "grant-equivalent" measurement, which would value concessional loans at their subsidy content rather than their face value, emerged as a technical device with enormous distributive implications: under face-value accounting, a $100 million concessional loan at near-market rates counted the same as a $100 million grant—a comparison that inflated donor contributions and understated recipient costs.

The accountability vocabulary crystallised in parallel, but through different institutional channels. Where "leverage" and "blended finance" emerged from OECD working papers and development bank reports, "additionality," "double counting," and "grant-equivalent" were forged in the space between civil society critique and academic political economy. Oxfam's first climate finance assessments (2012) reframed the $100 billion pledge as an accountability test, not a mobilisation target. Roberts and Weikmans (2017) and Michaelowa and Michaelowa (2007) gave the critique academic form: not merely questioning the *numbers* but challenging the *categories* through which they were produced. By 2014, the accountability pole had its

own lexicon, its own institutional base (NGO shadow reports, UNFCCC submissions from the G77+China), and its own metric (grant-equivalent valuation as counterpoint to face value). The two vocabularies were not independent inventions; they emerged in direct reaction to each other, each defining itself against the other's accounting conventions.

2.4 Economists as architects

A notable feature of this period is the role of individual economists in designing the policy architecture. The UN Secretary-General's High-Level Advisory Group on Climate Change Financing (AGF), convened in 2010, brought together a deliberately hybrid body: co-chaired by Prime Minister Meles Zenawi of Ethiopia and Prime Minister Jens Stoltenberg of Norway, it included finance ministers, central bankers, and economists alongside heads of state. Nicholas Stern, whose Review (Stern, 2007) had established the economic case for early action, served as a member, bridging the macro-economic framing of "how much will it cost?" with the institutional question of "through what channels?" (United Nations Secretary-General's High-level Advisory Group on Climate Change Financing, 2010). The AGF's central analytical innovation was a tripartite framework distinguishing "sources" (where money comes from), "instruments" (how it is channelled), and "governance" (who decides). This framework did not merely *describe* how $100 billion could be raised; it *constituted* the analytical categories through which the target was subsequently operationalised. One politically sensitive question—whether revenues from carbon pricing mechanisms such as financial transaction taxes or bunker fuel levies should count as "climate finance sources"—was left strategically ambiguous. The report concluded that the target was "challenging but feasible," a formulation that reverse-engineered the political commitment as achievable rather than assessing whether it was adequate. The sources/instruments/governance framework became standard in subsequent policy debates, illustrating how a small group's analytical choices can structure an entire field's categories.

Stern was not alone. Jean-Charles Hourcade, an economist at CIRED who had long argued that climate policy and development finance were inseparable, contributed to the intellectual architecture through a different channel: the argument that the transition to low-carbon economies required overcoming a "financial paradox"—abundant global savings coexisting with insufficient investment in climate-compatible infrastructure (Hourcade et al., 2015). Where Stern emphasised the cost-benefit case for early action, Hourcade emphasised the structural mismatch between the temporality of climate investment (long-term, infrastructure-heavy) and the short-termism of financial markets. This strand of thinking would later inform debates about "blended finance" and "de-risking," framing public climate finance not as aid but as a correction of financial market failures.

The point is not that any individual economist determined the architecture, but that economists as a group occupied a distinctive position at this juncture. Unlike diplomats, who negotiated political commitments, or climate scientists, who established physical baselines, economists provided the *quantitative grammar* through which commitments could be operationalised. They did not act alone: the *demand* for quantification came from G77+China

negotiators and civil-society organisations who insisted that pledges be measurable and verifiable, and the political targets themselves were set by heads of state. But it was economists in international organisations who built the measurement instruments—the markers, ratios, and reporting frameworks—that gave these demands institutional form. In doing so, they determined what would be visible and what would remain invisible in the accounting of climate finance.

2.5 Corpus evidence

The corpus vocabulary shifts sharply in 2007–2008—the years when the Bali Action Plan mandated measurable, reportable, and verifiable (MRV) financial commitments and the global financial crisis reframed public investment. The onset of crystallization coincides with a negotiation moment rather than an intellectual "discovery." The composition chart (Figure 2) shows that by 2013 the six thematic communities identified by clustering were all established. The CDM/Kyoto vocabulary of the first period contracted sharply; terms associated with mobilisation and green finance expanded in the lexical indicators. After 2007, the evolution was gradual consolidation, not a second rupture. What changed after 2013 was the *routinisation* of measurement practices: the OECD's DAC statistics incorporated climate markers as standard fields; the UNFCCC's biennial assessments became regular exercises; the CPI's *Global Landscape* was published annually. Climate finance had become, in the terminology of the sociology of quantification, a "statistical convention"—a set of categories whose artificiality had become less visible through repeated use (Desrosières, 1998).

The publication ecology of the most-cited works confirms the institutional character of the crystallization. Alongside conventional academic journals (*Climate Policy*, *Nature Climate Change*, *Climatic Change*), the core includes a substantial share of institutional publications. Category-making power was exercised not through peer-reviewed articles alone but through the "grey" channels of policy reports, expert group papers, and institutional working paper series. The OECD/IEA Climate Change Expert Group Papers provided the technical foundations for DAC reporting standards. The Climate Policy Initiative's annual *Global Landscape of Climate Finance* report (Büchner et al., 2013), first published in 2012, became the most widely cited single source on climate finance volumes. These publications acquired epistemic authority through a productive circularity: they were produced by the institutions responsible for *implementing* the categories they described—a circularity that reinforced the categories' stability but also insulated them from external challenge (Gieryn, 1983).

Critically, the most influential papers show remarkable thematic stability across the 2005–2020 period. The categories through which economists analysed climate finance—carbon markets, green bonds, adaptation, accountability—were established by the mid-2000s. The vocabulary shift visible in the full corpus is driven by the influx of new, lower-cited scholarship, not by a reorientation of influential works. This finding supports the thesis that crystallization was a categorical event: the concepts were set once and have structured all subsequent debate.

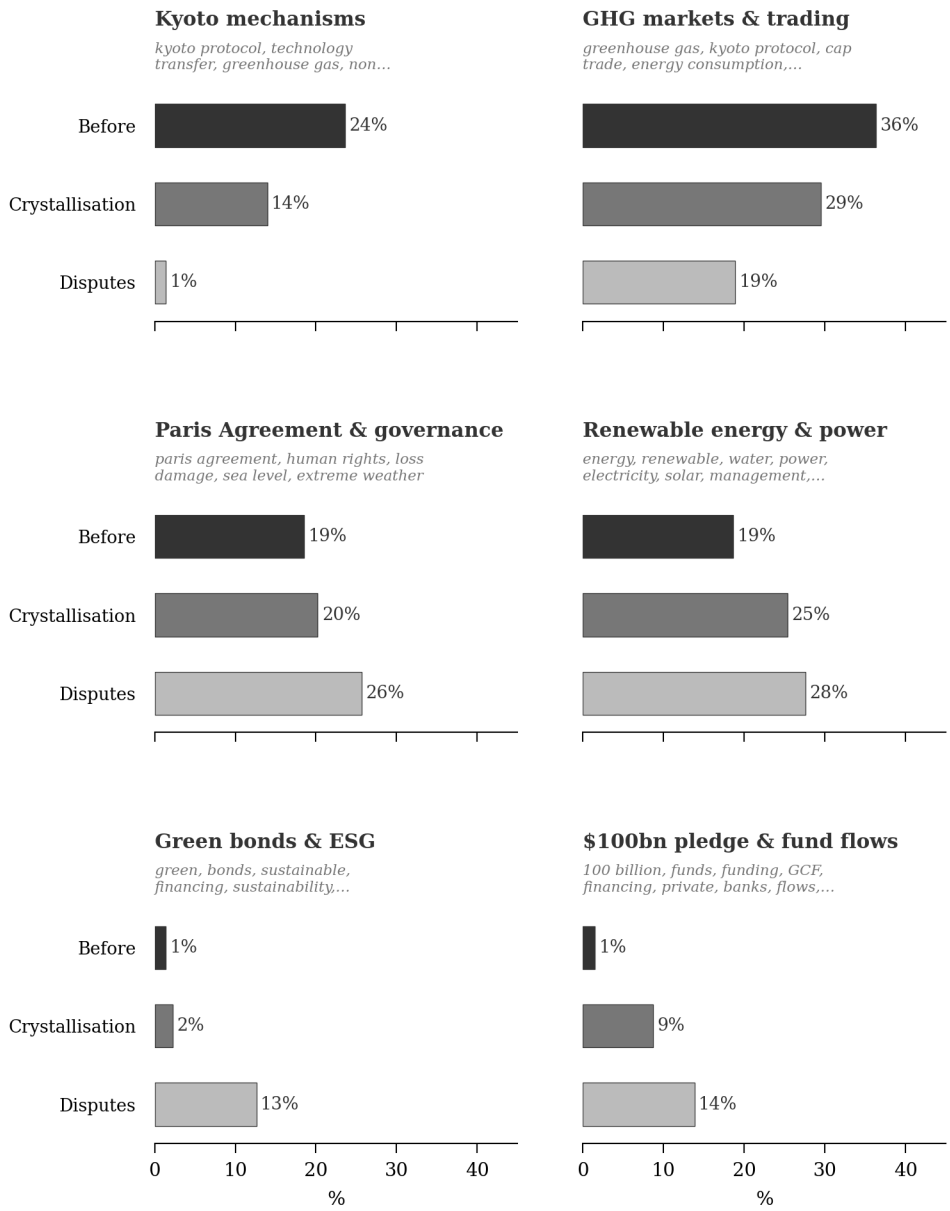

Figure 2: Thematic recomposition of climate finance scholarship across three periods. Six clusters were fitted by k-means (k=6) on text embeddings of the full corpus; each panel shows one cluster's share per period. Bold labels are author interpretations; italic terms are the top TF-IDF distinctive words. Kyoto mechanisms nearly vanish while GHG markets decline; green bonds and ESG, Paris Agreement governance, and renewable energy grow.

3. The Established Field (2015–2024)

By December 2015, climate finance had become a settled category of international law. The Paris Agreement's Article 9 codified the obligation in language that would have been unthinkable in 1992:

> Developed country Parties shall provide financial resources to assist developing country Parties with respect to both mitigation and adaptation in continuation of their existing obligations under the Convention. […] Such mobilization of climate finance should represent a progression beyond previous efforts. (United Nations, 2015, Article 9.1, 9.3)

Where Article 4.3 had specified "agreed full incremental costs" through a bilateral mechanism, and the Bali Action Plan had demanded "adequate, predictable and sustainable financial resources," Paris simply wrote "financial resources" as if the category required no further definition. The accounting infrastructure built during crystallization had done its work: "climate finance" now referred to something that everyone could name, even if no one could agree on how to measure it. The Paris Agreement and the Glasgow Climate Pact (2021) introduced new procedural demands—enhanced transparency frameworks, common reporting formats, a new collective quantified goal—but they did not redesign the conceptual infrastructure built during crystallization. The categories, metrics, and institutional arrangements described in Section 2 remained the scaffolding within which post-2015 debates unfolded. This does not mean that nothing changed; it means that the changes occurred *within* categories that had already been established.

3.1 Paris and the transparency framework

Beyond the financial obligation codified in Article 9, the Paris Agreement introduced several procedural innovations. The Enhanced Transparency Framework (ETF) established common reporting formats for tracking financial flows, replacing the ad hoc reporting of the pre-Paris era. And the agreement's emphasis on "nationally determined contributions" (NDCs) created new categories of finance—adaptation finance, loss-and-damage finance, technology transfer—each with its own measurement challenges.

Yet these innovations operated within the conceptual infrastructure built during crystallization. The ETF's common reporting formats, negotiated through the Subsidiary Body for Scientific and Technological Advice (SBSTA), drew directly on OECD DAC categories: the same instrument classifications (grants, concessional loans, non-concessional loans, equity, guarantees), the same channel distinctions (bilateral, multilateral), and the same sectoral codes. The methods for counting "mobilised" private finance followed the frameworks developed by Jachnik, Caruso, and Ellis at the OECD (Jachnik et al., 2015). Even the most contested element—whether to report at face value or grant-equivalent—was resolved by allowing both, deferring

rather than settling the methodological dispute. Pauw et al. (2022) argued that the post-2025 target negotiations revealed the governance consequences of this deferral: without agreed accounting modalities, the target became a number without a stable referent, its meaning dependent on which conventions each party chose to apply.

The basic distinction between public and private, bilateral and multilateral, grants and loans remained unchallenged. Paris refined and extended the measurement apparatus; it did not redesign it. Notably, other measurement regimes developed in parallel without displacing the OECD/DAC framework: the multilateral development banks adopted a joint methodology for tracking climate finance from 2011, using project-level activity-based accounting rather than the DAC's policy-marker approach; and the Green Climate Fund developed its own investment framework and results measurement system. These competing commensurations coexisted rather than converged, each producing different numbers from overlapping financial flows. The International Development Finance Club (IDFC), representing 27 national and regional development banks largely from the Global South, publishes its own annual Green Finance Mapping using categories that overlap with but do not replicate the OECD/DAC framework.

3.2 The four controversies after Paris

The post-Paris period did, however, intensify the four controversies identified in the introduction. Each became more technically sophisticated and more politically charged, but all remained anchored in the categories established during crystallization.

The valuation of concessional loans. The question of whether climate finance should be counted at face value or grant-equivalent became the subject of prolonged negotiation within the DAC. In 2014, the DAC agreed to adopt a new "grant-equivalent" methodology for measuring ODA, implemented in reporting from 2018 onward and replacing the previous face-value system. When applied to climate finance, the difference was dramatic: a $1 billion concessional loan with a 10% grant element would count as $1 billion under the old system but only $100 million under the new one. Donor countries that had been reporting large climate finance figures through loans saw their totals shrink. The technical change in measurement directly altered the political arithmetic of compliance with the $100 billion target (Bachus and Bécault, 2017). The Oxfam Climate Finance Shadow Reports, published annually since 2016, performed a distinctive form of *boundary work* (Gieryn, 1983): using the same data sources as the OECD but applying different accounting conventions to produce radically lower numbers, they consistently estimated the real transfer at roughly one-third of official figures (Carty and Le Comte, 2018; Zagema et al., 2023). The choice of metric was never neutral: face value says "the money exists"; grant-equivalent says "the subsidy exists." Each makes certain realities visible (aggregate flows, leverage multiples, donor effort) and others invisible (recipient needs, adaptation gaps, the quality of financed activities).

More broadly, climate finance categories function as *boundary objects* (Star and Griesemer, 1989): shared across communities with different interpretations while maintaining enough identity to coordinate action. The OECD–UNFCCC contest, the efficiency–accountability divide, and the Oxfam counter-expertise all illustrate *boundary work* (Gieryn, 1983) around these shared objects. Dalmedico (2010) showed that climate models function as boundary objects across science and governance; accounting categories operate similarly, enabling coordination precisely because their ambiguity permits different readings.

The credibility of Rio markers. Axel and Katharina Michaelowa's research programme on the quality of climate aid coding reached its most pointed conclusions in this period. Their studies documented systematic over-reporting: donor agencies had incentives to tag projects as climate-relevant regardless of their actual climate content, because the Rio markers were self-assessed and subject to no external verification (Michaelowa and Michaelowa, 2007; Weikmans and Roberts, 2019). A project to build a school might receive a "significant" climate marker if the building included rainwater harvesting, inflating the apparent volume of climate finance without changing the underlying reality. The markers, originally designed as a rough tracking tool, had become load-bearing elements of an international accounting system for which they were never intended. This is a case of what Power (1997) calls "audit failure"—not the absence of verification, but the transformation of a verification instrument into an incentive structure that undermines the very quality it was supposed to assure.

The attribution of mobilised private finance. The question of how to count private finance "mobilised" by public interventions grew more contested as the $100 billion deadline approached. Stadelmann et al. (2013) identified the core difficulty: private finance does not carry a label indicating which public intervention triggered it, so any attribution method involves counterfactual reasoning about what would have happened without the public contribution. Different methodologies yielded wildly different figures: the OECD's attribution approach, which counted only private finance directly linked to specific public interventions through documented causal chains, produced smaller numbers than the "total climate finance landscape" approach used by the Climate Policy Initiative, which included all private investment in climate-relevant sectors regardless of public involvement (Büchner et al., 2013). Atteridge and Dzebo (2015) showed that the Copenhagen pledge's ambiguous inclusion of private finance had created the measurement problem in the first place: the political commitment preceded the methodology, forcing economists to build attribution frameworks after the fact. The choice of methodology was not neutral—it determined whether developed countries could claim to have met their Copenhagen commitment.

The boundary between development assistance and climate obligation. Roberts and Weikmans (2017) argued that the recurring accounting disputes were symptoms of a deeper political conflict over the boundary between development assistance and climate-specific obligations. Developed countries had an interest in counting as much of their existing aid as possible as "climate finance," thereby meeting the $100 billion target without increasing total financial transfers. Developing countries, conversely, argued that climate finance should be "new and additional" to existing aid commitments—a principle enshrined in the Copenhagen Accord

but never operationally defined. Stadelmann et al. (2011) had shown that eight different baseline options existed for assessing additionality, and only two met both feasibility and equity criteria; governments chose none of them. Skovgaard (2017) demonstrated that finance ministries in donor countries framed climate finance as a subset of existing ODA in order to avoid new budgetary commitments, revealing how institutional interests shaped the very definition of the category. The boundary remained contested because the very categories used to draw it (ODA, concessionality, additionality) were products of the development finance system, not of climate policy. So, increasingly, was the donor–recipient binary itself: China, the Gulf states, and other upper-middle-income non-Annex I countries complicated the 1992 division between providers and recipients of finance. The Paris Agreement's formulation "in light of different national circumstances" was a deliberate retreat from binary classification, and the NCQG negotiations at Baku (2024) reopened the question of whether major emerging economies should contribute—without resolving which economic criterion might replace the frozen Annex lists. Steckel et al. (2016) proposed reframing the debate entirely, arguing for a shift "from climate finance toward sustainable development finance" that would dissolve the boundary rather than police it. This proposal gained little traction; the existing categories were already too institutionally entrenched to dissolve.

3.3 The $100 billion claim and the $300 billion transition

In 2022, the OECD announced that developed countries had, for the first time, exceeded the $100 billion annual target—reaching $115.9 billion. The claim rested entirely on the statistical infrastructure built during crystallization: DAC reporting categories, Rio markers, mobilised-private-finance methodologies. But the number was immediately contested. The Oxfam Climate Finance Shadow Report (Carty and Le Comte, 2018), published annually since 2016, applied three systematic adjustments: counting loans at grant-equivalent rather than face value, discounting projects with questionable climate relevance (based on independent spot-checks of Rio marker coding), and excluding non-concessional instruments. Under these corrections, Oxfam estimated the "climate-specific net assistance" at roughly $21–$24 billion in the years when OECD reported $60–$80 billion—less than a third of the official total. The Shadow Reports represented a form of *counter-expertise*: civil society economists using the same data sources but different accounting conventions to produce radically different numbers. The dispute was not about facts but about which commensuration was legitimate (Forstater, 2012).

At COP27 in Sharm el-Sheikh (2022), parties agreed to establish a fund for loss and damage—the first formal acknowledgment that some climate impacts cannot be adapted to and must be compensated. This category potentially disrupts the crystallization-era framework: loss-and-damage finance is neither mitigation nor adaptation, does not fit neatly into the ODA categories, and raises questions of liability that the existing accounting system was designed to avoid. Whether it will generate its own measurement infrastructure or be absorbed into the

existing categories remains an open question, but it represents the most significant challenge to the conceptual architecture established during crystallization.

At COP29 in Baku (November 2024), parties agreed on a New Collective Quantified Goal (NCQG) of $300 billion per year by 2035. The negotiations revealed the same categorical tensions, now amplified. Which financial instruments would count? Would South-South contributions be included? China's bilateral development finance in Africa—channelled through China Development Bank and the Forum on China-Africa Cooperation—is the most prominent example: these flows are not reported through the OECD DAC system, making them invisible to the accounting infrastructure described in §2, and whether they count as "climate finance" depends entirely on which measurement framework is applied. How would currency fluctuations and inflation be handled? The answers to these questions depended not on new theory but on the accounting categories stabilised two decades earlier. The NCQG was fought—like the $100 billion before it—within the intellectual infrastructure that economists had built during the crystallization period.

3.4 Corpus evidence

The $100 billion dispute and the NCQG negotiations were fought within categories established during crystallization. The corpus confirms this: neither the Paris Agreement (2015) nor the Glasgow Climate Pact (2021) produces a vocabulary shift comparable to the 2007–2008 recomposition. A qualification is essential: this finding concerns *vocabulary and measurement categories*, not institutional architecture. The Enhanced Transparency Framework moved climate finance reporting from voluntary to mandatory; the TCFD (2017) extended climate-finance categories into the domain of financial-risk regulation—a genuinely new application not prefigured during crystallization. What persisted was the conceptual infrastructure: the categories through which flows are classified, counted, and contested. The thematic panels (Figure 2) decompose the corpus into six clusters fitted to the full 1990–2024 period by unsupervised clustering (k-means on text embeddings of approximately 30,000 works). The decomposition primarily reflects the post-2015 literature, which constitutes over two-thirds of publications; per-period clustering reveals that some themes—land use, clean development, emissions trading—appear across all three periods, while others are period-specific (green bonds emerge only after 2015). The largest shifts are the sharp decline of CDM and renewable energy and the dramatic growth of green bonds and ESG. No cluster disappears entirely, but the six-cluster frame is a descriptive device for tracking thematic recomposition, not a claim that six natural communities independently emerge in each period.

The OECD–Oxfam dispute described above is the direct expression of a deeper structural feature. A persistent tension runs through the literature between two poles: an "efficiency" pole characterised by leverage, de-risking, blended finance, and green bonds, and an "accountability" pole characterised by additionality, climate justice, grant-equivalent, and double counting. This two-pole structure is visible across both thematic and lexical analyses and remains detectable—though attenuated—in the most-cited core works. During crystallization, the two

orientations coexisted without sharp separation; in the post-2015 period, the divide sharpened markedly. The field did not fragment into unrelated communities; it polarised along an axis that was already latent in the categories established during crystallization.

The two poles are institutionally embedded. The efficiency pole publishes in finance and energy journals (*Review of Financial Studies*, *Journal of Financial Economics*, *Energy Economics*); the accountability pole publishes in climate governance and environmental policy venues (*Global Environmental Politics*, *Climatic Change*, *WIREs Climate Change*). A few journals—notably *Climate Policy*—serve as shared ground (Table 2). The venue ecology confirms that the divide is not an artifact of the computational method: it reflects where researchers choose to publish, and therefore which communities they address.

The growth was asymmetric. Publications near the efficiency pole—leverage, blended finance, de-risking, climate risk (Monasterolo, 2020)—multiplied faster than those near the accountability pole, shifting the field's centre of gravity. But the accountability pole did not thin out. It diversified: new sub-literatures on loss and damage, climate debt, and just transition (Khan et al., 2019) replaced the earlier focus on additionality testing. The distribution showed two distinct concentrations in each period rather than a single shifting mass. What changed was the relative density, not the structure.

The tension between the two poles is constitutive: they sustain each other because they share the same categories even as they contest their interpretation. OECD reports frame climate finance as a tool for "mobilising" and "leveraging" private capital; Oxfam's assessments define their purpose as "holding donors to account" for the adequacy and quality of pledges. The tension is productive rather than pathological. If either pole prevailed—if climate finance became purely a financial engineering problem or purely a justice claim—it would lose its capacity to sustain both technocratic management and political accountability simultaneously. The two-pole structure detected in our corpus is the trace of this constitutive tension in the structure of scholarship itself.

Table 2: Publication venues of core works (cited ≥ 50) by pole assignment. Each work is assigned to the efficiency or accountability pole based on its position along the embedding axis (§3.4). "Lean" indicates the venue's overall orientation: journals where a majority of core papers fall on one side.

Lean	Journal	Eff.	Acc.	Total
Efficiency	Energy Economics	14	4	18
Efficiency	Sustainability	79	27	106
Efficiency	Renewable and Sustainable Energy Reviews	31	11	42
Efficiency	Energies	30	13	43
Efficiency	Energy Policy	46	21	67
Efficiency	Journal of Cleaner Production	25	15	40
Shared	Climate Policy	24	33	57
Shared	Ecology and Society	14	20	34
Accountability	Cambridge University Press eBooks	10	15	25
Accountability	Nature Climate Change	10	19	29
Accountability	International Environmental Agreements Politics Law and Economics	6	16	22
Accountability	Climatic Change	10	33	43
Accountability	Nature Communications	8	27	35
Accountability	Wiley Interdisciplinary Reviews Climate Change	7	28	35
Accountability	Global Environmental Politics	1	14	15

Conclusion

This paper has traced how climate finance was constituted as an economic object between 1990 and 2024—not through progressive discovery of a pre-existing reality, but through the construction of accounting categories (Rio markers, mobilised-private-finance methodologies, concessionality thresholds), the crystallization of these categories in a narrow window between 2007 and 2014, and the persistence of a constitutive tension between efficiency-oriented and accountability-oriented approaches that has structured the field ever since. The economists who shaped climate finance did not predict the future; they built the accounting infrastructure through which the present is assessed. Their contribution belongs less to the history of economic theory than to the history of economic governance—the production of commensuration devices through which states and international organisations render financial flows calculable and actionable.

But *from what* did they build? The legal foundation of climate finance was Article 4.3 of the UNFCCC (United Nations, 1992), which established a precise accounting logic: developed countries would reimburse the "agreed full incremental costs" incurred by developing countries in meeting their Convention obligations, through a bilateral mechanism between the Party and the Convention's financial entity. The framework was transactional, not aggregative. It specified which costs were covered, required bilateral agreement, and rested on a coherent economic concept—incrementality—with clear theoretical foundations in environmental economics.

What happened next was not an extension of this framework but its progressive abandonment. The Kyoto Protocol (1997) introduced market mechanisms (the CDM, Joint Implementation) that brought private flows and carbon credits into the picture, loosening the bilateral reimbursement logic. The Marrakech Accords (2001) institutionalised dedicated climate funds (the GEF, the Adaptation Fund), creating channels that operated alongside but not within the Article 4.3 framework. Then Copenhagen (2009) produced the decisive rupture: a $100 billion annual pledge that specified neither which costs were covered nor what "climate finance" meant, only that the money must be mobilised. The Paris Agreement's Article 9 (2015) completed the transformation by requiring developed countries to "mobilise" finance including private flows, a formulation that bears no resemblance to the bilateral reimbursement of agreed incremental costs envisaged in 1992. The NCQG agreed at Baku in 2024, $300 billion per year by 2035, was negotiated with the definition of "climate finance" still unresolved, three decades after the Convention.

The economists who built the measurement infrastructure did not simply fail to define their object. They *abandoned* a rigorous definition that was available—the incremental-cost framework of Article 4.3—because ambiguity proved more productive than precision. A precise definition would have constrained the countable total, limiting what donors could claim and what recipients could demand. The deliberately vague category served opposing interests simultaneously: donors could count concessional loans, guarantees, and "mobilised" private investment; recipients could argue that even these inflated totals fell short. The productive tension between efficiency and accountability that our corpus detects as a two-pole structure

is the scholarly trace of this strategic indeterminacy. Climate finance became a governable object not *despite* but *through* the theoretical incoherence of the category—an incoherence that was not a deficiency to be repaired but a political resource to be exploited.

Several limitations point to further research. Our multilingual corpus remains dominated by English; the intellectual history of climate finance as seen from recipient countries is largely unwritten—and the developed/developing binary itself, frozen in 1992 and never updated despite dramatic shifts in the global economy, remains one of the most consequential yet least examined constructions in climate finance governance. Our account of economists as institutional architects rests on published outputs rather than archival records or interviews; claims about individual motivations remain conjectural until primary-source research is undertaken. Our corpus covers academic and institutional publications; the role of professional financial media (*The Economist*, *Financial Times*, Bloomberg) in popularising and framing climate finance categories falls outside its scope. The trajectory from Article 4.3 to the NCQG itself deserves a dedicated legal-institutional history drawing on negotiation records and committee minutes. Whether the next generation of economists will return to the precision that Article 4.3 offered, or whether the productive ambiguity will prove durable, remains the open question of climate finance governance.

Data and code availability. The corpus, analysis scripts, and reproducible pipeline used in this study are available at [repository URL removed for review].

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